

L (09)

Topic :- Amines

Date :- 08/06 - Last part.

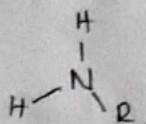
Naming Amines

- alkyl - substituted (alkylamines) or aryl - substituted (arylamines)
- classified.

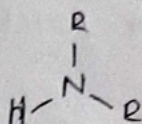
→ 1° ($R-NH_2$) primary

→ 2° (R_2NH) Secondary

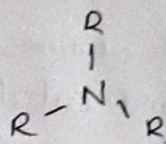
→ 3° (R_3N) Tertiary



Primary.



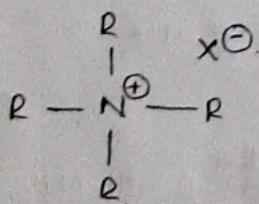
Secondary



Tertiary.

Quaternary Ammonium Ions

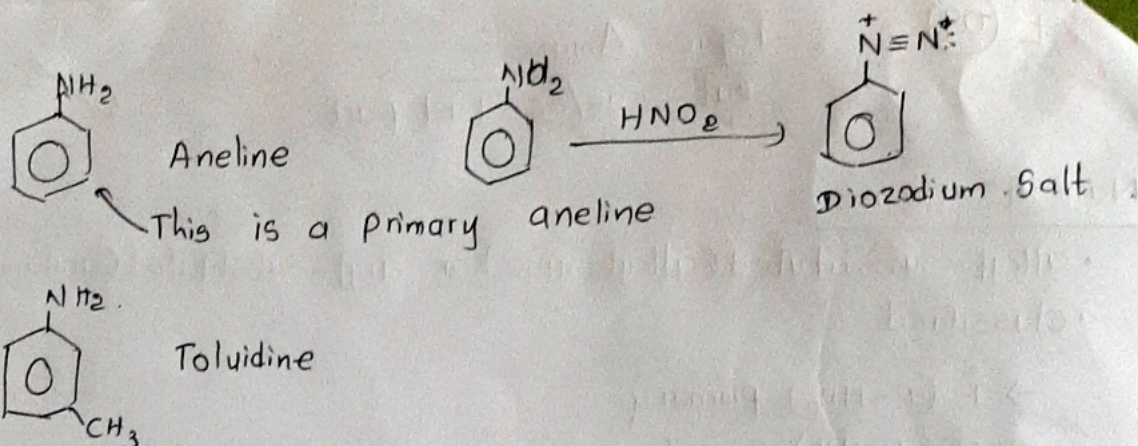
- A nitrogen atom with four attached groups is positively charged.
- Compounds are quaternary ammonium salts.



Common Names

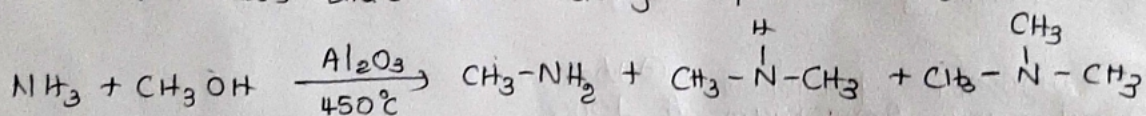
- alkylamines do not have common names unless they are biological molecules
- Simple arylamines have common names

Pseudo Science



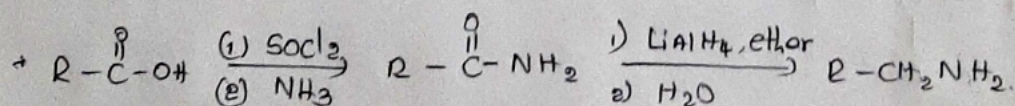
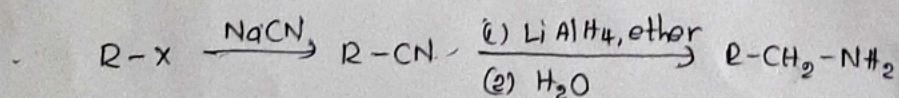
Properties and Sources of Amines

- Simple methylated amines from reaction of NH_3 with CH_3OH and alumina catalyst
- Yields a mixture of monomethylated, dimethylated and trimethylated products that are easily separated by distillation.



Synthesis of Amines

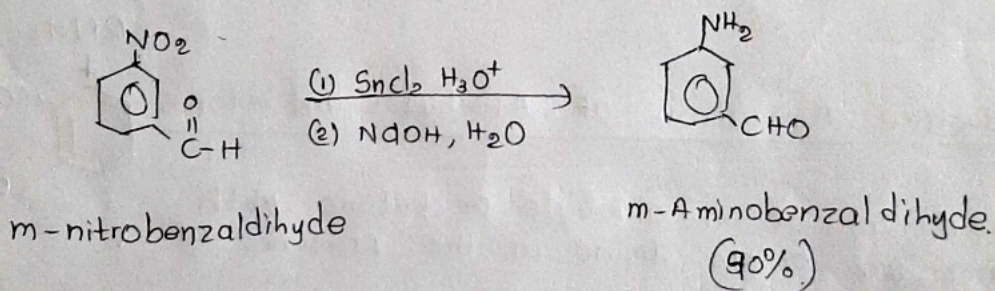
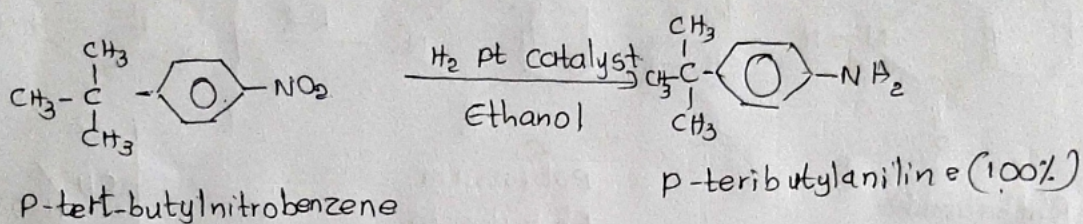
* Reduction of nitriles and amides



Reduction Aryl Nitro Compounds

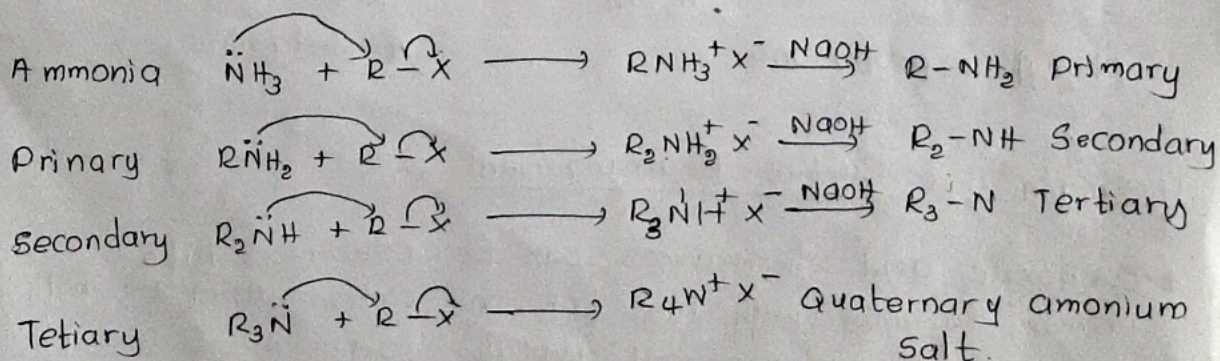
[02]

- Arylamines are prepared from nitration of an aromatic Compound and Reduction of nitro group
- Reduction by catalytic hydrogenation over platinum is suitable if no other groups can be reduced.
- Iron, Zn, tin and tin(II) chloride are effective in acidic solution.



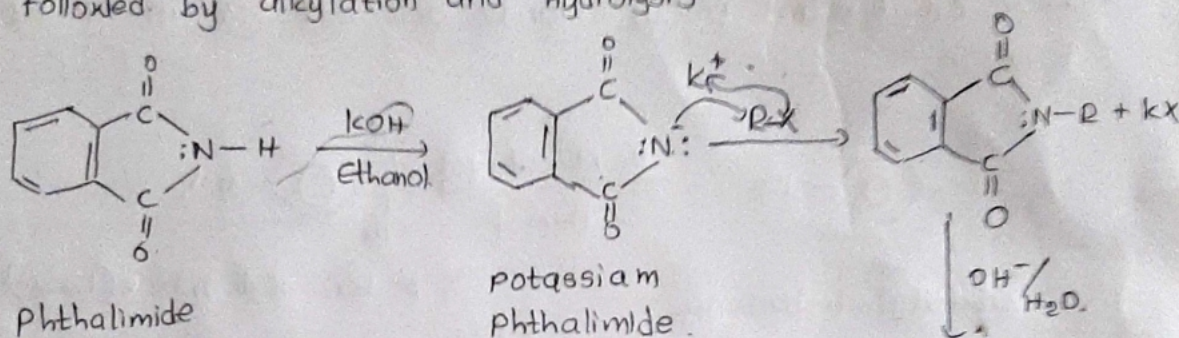
Reactions of Alkyl Halides

- Ammonia and other amines are good ~~nucle~~ Nucleophiles



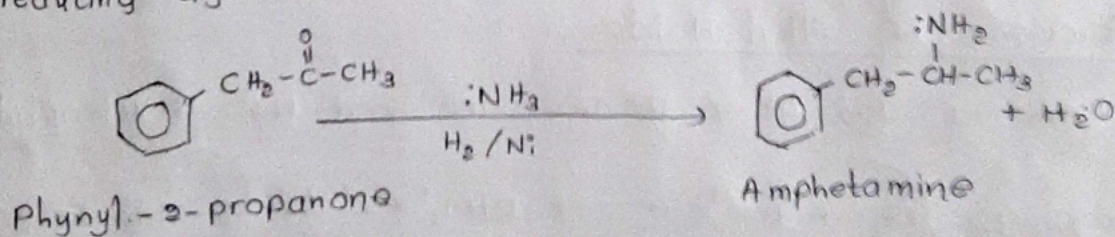
Gabriel Synthesis of primary Amines

- A phthalimide alkylation for preparing a primary amine from an alkyl halide
- The N-H imides ($-\text{CONHCO}-$) can be removed by KOH followed by alkylation and hydrolysis



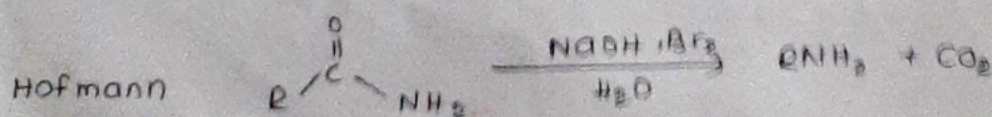
Reductive Amination of Aldehydes and ketones

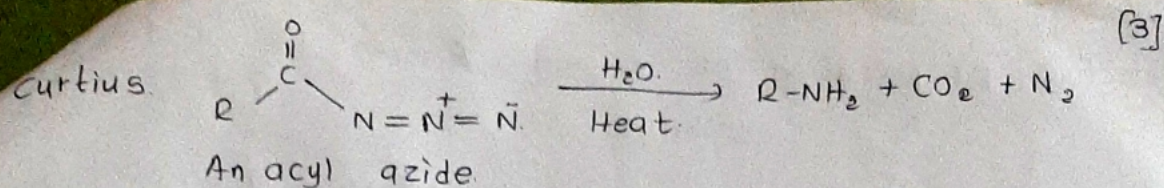
- Treatment of an aldehyde or ketone with ammonia or an amine in the presence of a reducing agent.



Hofmann and Curtius Rearrangement

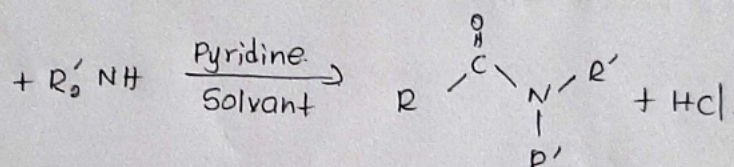
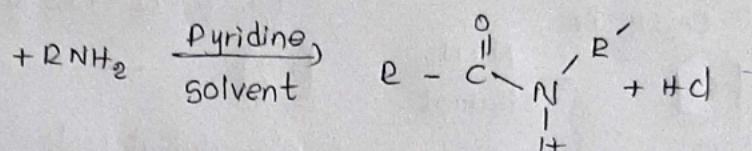
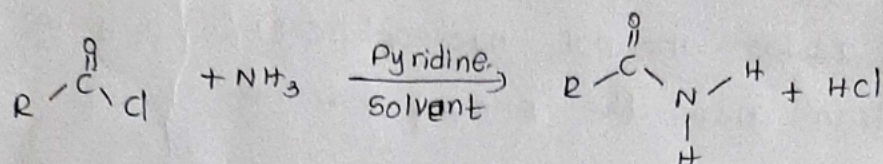
- * Carboxylic acid derivatives can be converted into primary amines with loss of one carbon atom by both the Hofmann rearrangement and the Curtius rearrangement.





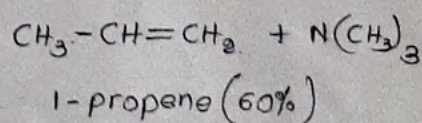
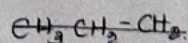
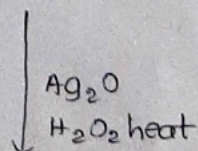
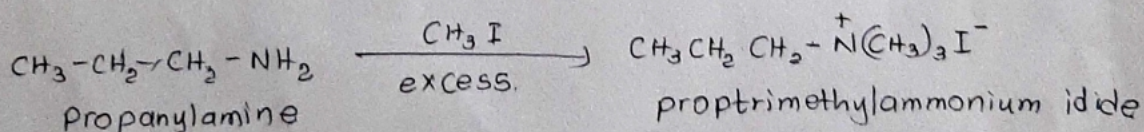
Reactions of Amines

- Alkylation and acylation have already been presented.



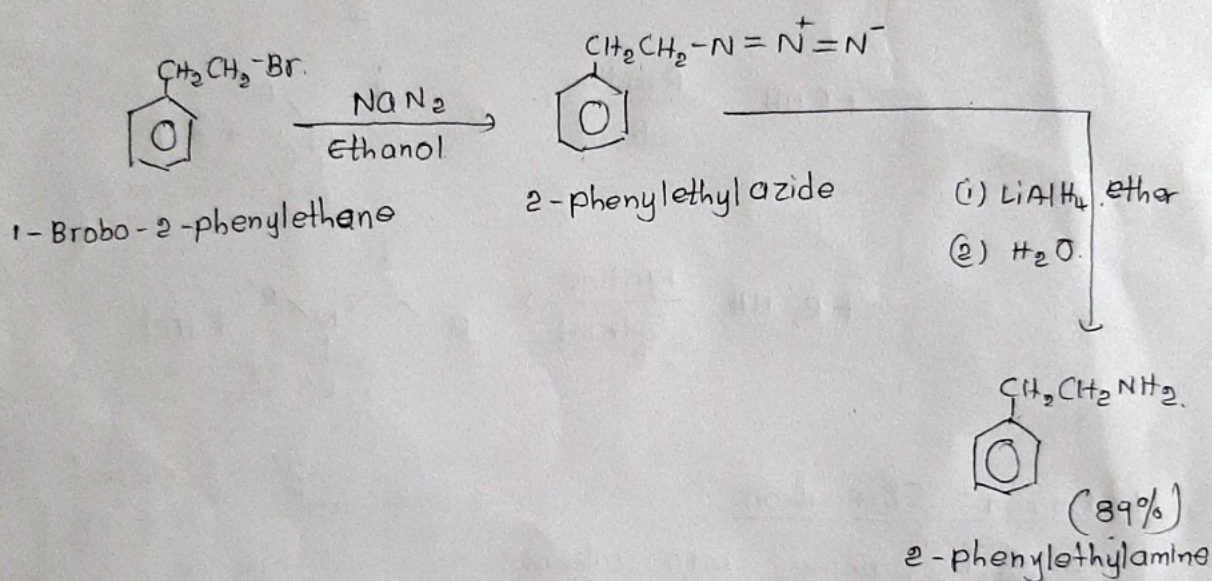
Hofmann Elimination

- Converts amines into alkenes
- NH_3^+ is very a poor leaving group so it converted to an alkylammonium ion, which is a good leaving group.



Selective preparation of primary Amines: the Azide Synthesis

- Azide ion, N_3^- displaces a halide ion from a primary or secondary alkyl halide to give an alkyl azide RN_3 .
- Alkyl azides are not nucleophilic (but they are explosive)
- Reduction gives the primary amine



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